which would only be slightly radioactive. The resulting scrap can either be encapsulated in concrete and put in a repository for normal reactor waste (Swedish plans), or melted to be used as raw material for new reactor parts where the remaining level of activity is acceptable (France).

20.5. SAFE OPERATION OF NUCLEAR REACTORS

The major risks from reactor operation stem from operator mistakes (<10%) or failure of critical hardware (>90%). The risk from human errors is reduced by recurrent training of personnel, especially persons working in the control room, using mockups and simulators. Thereby personnel in critical functions train their behaviour in normal situations, but also in situations which seldom or never occur during routine operation of the reactor. Also exchange of experience and the results from analysis of actions violating or threatening to violate safety instructions help to reduce the risk for human errors. Risks from hardware failures or malfunctions are minimized by a continuous work aiming to identify risk dominating sequences and critical hardware. Once identified, such sequences should be modified, procedures changed, equipment replaced or duplicated in order to reduce the overall risk until it reaches an acceptable level. It is important to remember that simple duplication will not reduce the risk from common mode failures very much.

Figure 20.9 shows an example of an *event tree* starting with an *initiating event* to the left and branching each time an action takes place. The upward branch corresponds to a success of the action specified at the top, whereas the downward branch represents a failure. Each branch is normally the compounded effect of actions in a sub-tree describing the function and interaction of affected hardware. The figures at each downward branch are estimated probabilities for a failure of the action at the top. The line representing the outcome of an event is then followed to the right; either to a final state or to another branch. To the extreme right is the final state with its estimated probability. For example, the most important risk in Fig. 20.9 arises from failure to restore AC power within 20 minutes; probability $(1/25) \times 10^{-6} = 4 \times 10^{-8}$ per reactor year. This was the main reason for the Fukoshima accident where the auxiliary diesel generators were drowned by the tsunami. For independent outcomes of events in a branch (e.g. probability of failure of too many of the available safety valves together with failing operation of the blow-down pipes), the probability is the product of the probabilities at the branches taken in the sub-tree. In case of common mode failures, the compounded probability is the same as the probability of a single event of that type. The probabilistic event tree is an example of the type of tools used to identify the current risk dominating event chain. When carried out meticulously, operator training, experience feedback and risk analysis help to continually improve the safety of reactor operation.

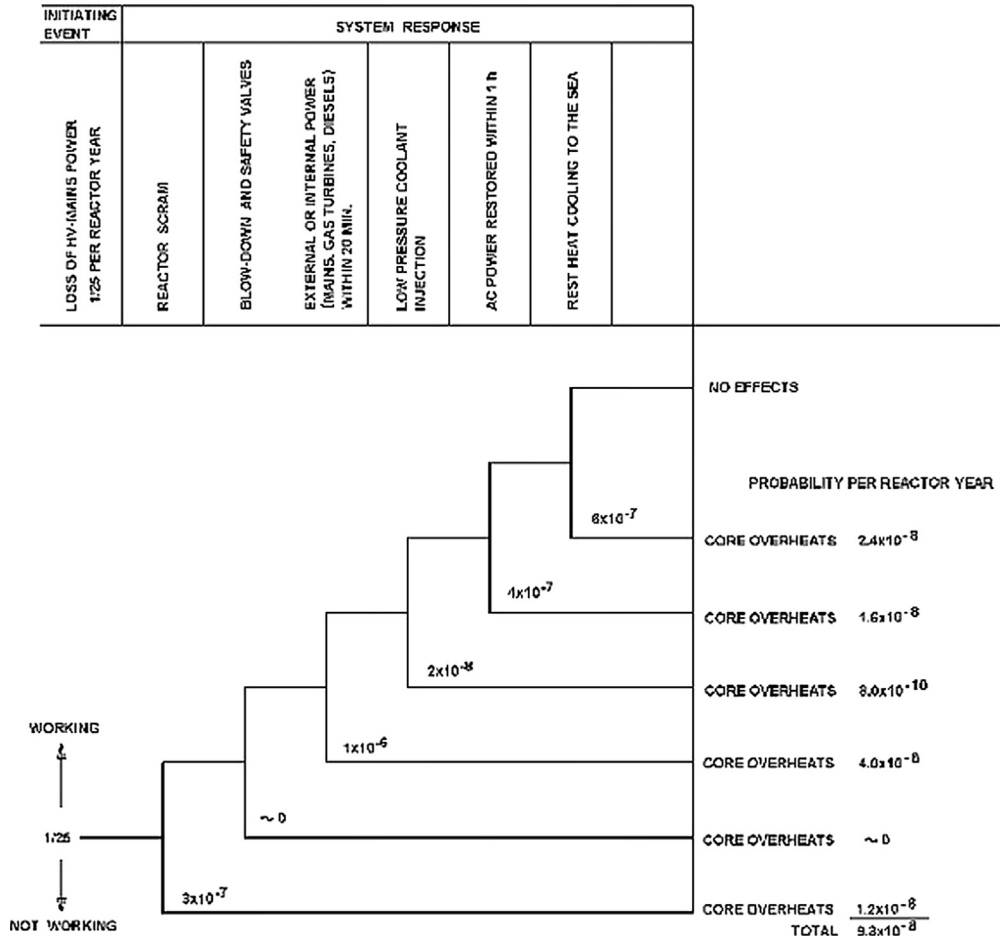


Figure 20.9 Example of an event tree for a hypothetical BWR.

20.6. EXERCISES

20.1. The gas space (volume 6000 m^3) in the PS system of a large BWR is filled with N_2 at 97 kPa and 25°C. Assume the same temperature in the dry and wet wells and that the heat of evaporation of water and its heat capacitivity are ~2.26 MJ/kg and ~4.18 kJ kg^{-1} $°C^{-1}$, respectively, up to ~100°C. The 2000 m^3 water in the condensation pool is at 25°C. The primary system contains steam at 287°C and 7 MPa (energy content ~2.8 MJ/kg steam). How many kg of such steam could be dumped into the condensation pool before the pressure in the PS system exceeds 0.1 MPa?